



DSN AI Bootcamp 2026 Project Participation – Data Analytics Track

Answer & Submission Sheet

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Dashboard tool used	Excel

Part A – Data Cleaning Review

Identify at least three meaningful data-quality issues you found. For each issue, explain what you found, what you did, and why your action was appropriate.

Data-quality issue	What you found	What you did	Why this action was appropriate
Duplicate transactions	The dataset contains 432 rows but only 420 unique transaction ids, i.e., 12 were duplicates	I identified duplicate transaction ids after cross-checking with other columns and removed duplicate copies	The duplicated rows were exact copies and represented the same transactions. Keeping them would double-count sales and distort revenue, profit, units, and other performance measures.
Inconsistent market/state labels	The same market appeared under different spellings/formats, including Lagos, Lagos and LAGOS.	I standardized the inconsistent labels into a single Lagos category.	These records represented the same market. Standardizing the labels prevents one market from being split into multiple categories and gives a more accurate market-level analysis.

Missing product cost values	31 cleaned transaction records had missing “Cost Per Unit” values. The observed unit cost for each affected product was otherwise consistent in the dataset.	I used the consistent observed product-level unit cost to fill the missing cost values and flagged the affected records as imputed.	The product costs showed a consistent pattern within each product. Imputing from the observed product cost allowed profit to be calculated while keeping the affected records identifiable rather than silently treating the missing values as zero.
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Which data-quality issue had the greatest impact on your analysis?

The incorrect/inconsistent Revenue values had the greatest impact on the final analysis.

Explain how correcting that issue changed or affected the analysis.

After removing duplicate transactions, I independently recalculated revenue using $\text{Units} \times \text{Unit Price} \times (1 - \text{Discount})$. The reported Revenue did not reconcile with this calculation for 22 of the 420 unique transactions. The deduplicated reported revenue was ₦46,369,970, while the corrected calculated revenue was ₦46,679,725, a difference of ₦309,755. This affected the reliability of revenue and profit analysis and showed that the reported Revenue field could not be used directly without validation.

Part B – Dashboard Audit

For each management claim, state your verdict, provide the corrected result where applicable, and explain the evidence supporting your conclusion.

1. Abuja is the most profitable market.

Verdict

Incorrect

Corrected answer / result

Lagos is the most profitable market, with corrected profit of ₦4,579,900. Abuja recorded ₦1,698,100.

Explanation / evidence

After cleaning the market labels and correcting the transaction-level calculations, Lagos had the highest profit among the five markets. Abuja had the lowest corrected profit and therefore the original conclusion that Abuja was the most profitable market was not supported by the corrected data.

2. Electronics is the strongest-performing product category and should receive more budget.

Verdict



Misleading

Corrected answer / result

Electronics generated the highest revenue and profit, with **₦29,346,500 revenue** and **₦8,986,500 profit**. However, Accessories had the highest profit margin at **46.76%**, compared with **30.62% for Electronics**, and also had fewer returns.

Explanation / evidence

Electronics is the largest category by total revenue and profit, but the recommendation to automatically allocate more budget to it does not follow from revenue alone. Accessories generated a smaller total contribution but had a substantially higher margin and lower return count. The dataset therefore supports further evaluation rather than a simple "increase Electronics budget" conclusion.

3. Higher discounting is improving profitability.

Verdict

Incorrect

Corrected answer / result

Profit margin decreased as the observed discount level increased. Margins ranged from **35.69% at 0% discount** to **15.29% at 25% discount**.

Explanation / evidence

The corrected analysis does not support the claim that higher discounting improves profitability. The observed margins were 35.69% at 0%, 33.01% at 5%, 28.45% at 10%, 27.14% at 15%, 20.79% at 20%, and 15.29% at 25%. This is an observed association in the dataset and does not by itself establish causation.

4. The Online channel has the lowest return rate.

Verdict

Incorrect

Corrected answer / result

The **In-store** channel had the lowest return rate at **6.44%**. Online was **9.70%**, while WhatsApp was **11.90%**.

Explanation / evidence

The corrected return analysis shows that Online was not the lowest-return channel. In-store had the lowest return rate, while WhatsApp had the highest among the three channels.

5. August is the strongest sales month.

Verdict

Incorrect

Corrected answer / result

May was the strongest sales month, with corrected revenue of **₦7,008,500**. August was the weakest, with **₦3,167,425**.



Explanation / evidence

After removing duplicate transactions and validating revenue, monthly revenue showed May as the highest month. August recorded substantially lower revenue than every other month in the dataset

6. Lagos is underperforming compared with other markets.

Verdict

Incorrect

Corrected answer / result

Lagos was the strongest market by corrected revenue and profit, generating **₦13,692,700 revenue** and **₦4,579,900 profit**.

Explanation / evidence

After consolidating the inconsistent Lagos labels, all Lagos records were analyzed as one market. Lagos had the highest corrected revenue and profit and a return rate of approximately **5.65%**, so the evidence does not support the conclusion that Lagos was underperforming.

7. The reported Revenue field can be used directly for decision-making.

Verdict

Incorrect

Corrected answer / result

The reported Revenue field should not be used directly without validation. **22 of the 420 unique transactions** did not reconcile with the independently calculated revenue.

Explanation / evidence

I recalculated revenue using $\text{Units} \times \text{Unit Price} \times (1 - \text{Discount})$. The reported revenue for 22 transactions differed from the independently calculated values. Across these mismatches, the reported revenue was **₦309,755 lower** than the corrected calculated revenue. Therefore, the Revenue field requires validation before being used for management decisions.

8. The dashboard is sufficiently reliable for management to act on without further data cleaning.

Verdict

Incorrect

Corrected answer / result

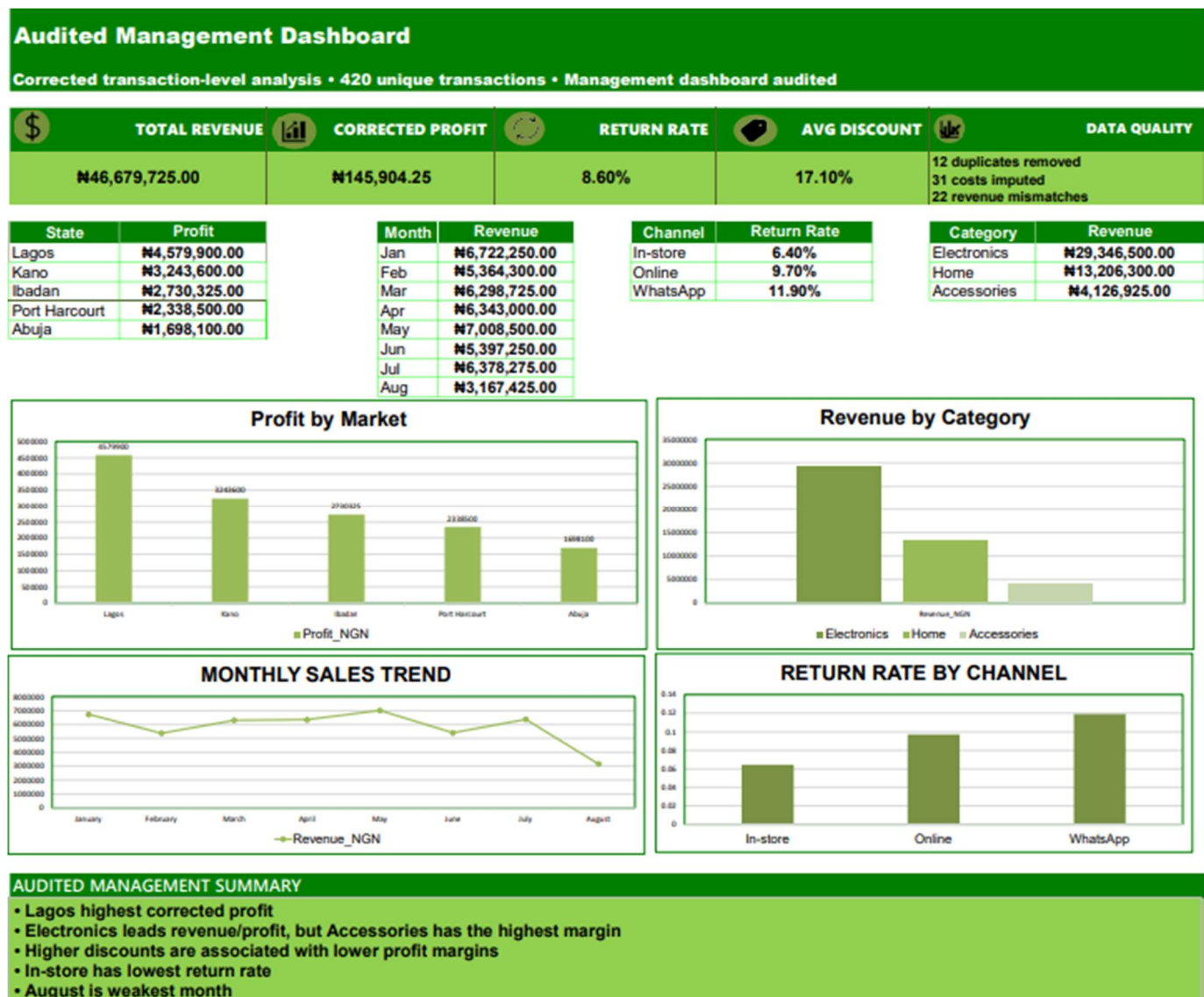
The dashboard required further cleaning and validation before being used for management decisions.

Explanation / evidence

The dataset contained 12 duplicate transaction records, inconsistent market/category labels, 31 missing product-cost values, and 22 revenue mismatches among the 420 unique transactions. These issues could affect revenue, profit, market, category, and other performance conclusions. The corrected dashboard therefore provides a more reliable basis for analysis than the original dashboard.

Part C – Corrected Dashboard

Insert a clear snapshot of your corrected dashboard below. The dashboard may be created in Power BI or Excel.





Part D – Final Analysis Questions

1. Which conclusion in the original dashboard was the most misleading, and why?

Answer

The conclusion that higher discounting was improving profitability was the most misleading.

Explanation / evidence

The corrected analysis showed the opposite pattern in the observed data. Profit margin declined from **35.69% at 0% discount** to **15.29% at 25% discount**. Because discounts reduce the effective selling price, the higher-discount groups showed lower margins. The original conclusion could therefore encourage management to use deeper discounts without evidence that this would improve profitability.

2. What changed most materially after you cleaned and corrected the data?

Answer

The reliability of the revenue and profit analysis changed most materially.

Explanation / evidence

After removing 12 duplicate records and recalculating revenue, reported revenue changed from **₦46,369,970** on the deduplicated records to **₦46,679,725** based on the validated transaction-level calculation. The corrected profit was **₦14,590,425** after addressing the missing cost values. Cleaning also changed how markets and categories were grouped because inconsistent labels were consolidated.

3. If management had acted on the original dashboard, what decision could have created the greatest business risk?

Answer

A major risk would have been making budget and pricing decisions based on unvalidated revenue and the conclusion that higher discounting improves profitability.

Explanation / evidence

The original dashboard could encourage management to increase discounting or allocate resources based on revenue figures that did not fully reconcile with the underlying transaction data. The corrected analysis shows that profit margins declined as observed discount levels increased, while 22 transactions also contained revenue discrepancies. This could lead to reduced margins and poorly targeted resource allocation.

4. Based on your corrected analysis, what is the most important driver of business performance in this dataset?

Answer

The most important driver visible in this dataset is the relationship between **sales volume/revenue and profitability**, particularly the effect of discounting on margin.

Explanation / evidence



Electronics generated the largest total revenue and profit, showing the importance of category sales volume. However, the discount analysis showed that higher observed discount levels were associated with progressively lower profit margins. This means that generating revenue alone is not sufficient; the profitability of that revenue is also important. The dataset therefore highlights the need to balance sales volume with mmargin protection.

5. What three actions should management take next based on the evidence from your corrected dashboard?

Answer

1. Review discounting strategy and protect profit margins.
2. Investigate market and category performance before reallocating budget.
3. Strengthen data-quality controls before future decision-making.

Explanation / evidence

1. **Review discounting strategy and protect profit margins.**
Avoid assuming that deeper discounts automatically improve performance. Review discount levels by product, market, and channel and monitor their effect on margin.
2. **Investigate market and category performance before reallocating budget.**
Electronics should not be evaluated on revenue alone. Management should compare revenue, profit, margin, and returns when deciding where additional resources should go.
3. **Strengthen data-quality controls before future decision-making.**
Introduce validation checks for duplicate transactions, inconsistent labels, missing costs, and revenue calculations so that management dashboards are based on reliable transaction-level data

Submission reminder: Complete all sections, insert a readable corrected-dashboard snapshot, ensure your written figures match the dashboard, and keep your Power BI or Excel working file available if verification is requested.